

Tong Statistical Field Theory — Sheet 2: Correlations and Scaling

D. Tong, Statistical Field Theory, Example Sheet 2; 6 worked problems.

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Contents

1	Sheet 2, Problem 1: Ornstein–Zernicke Correlations	1
2	Sheet 2, Problem 2: The Two-Point Function as a Green Function	2
3	Sheet 2, Problem 5*: Scaling in a Lifshitz Theory	2
4	Sheet 2, Problem 6: Scalar–Gauge Coupling	3
5	Sheet 2, Problem 7: Emergent Rotational Symmetry	3
6	Sheet 2, Problem 8: Wick’s Identity for a Gaussian Ensemble	3

These prompts paraphrase David Tong’s Statistical Field Theory, Example Sheet 2 (November 2018). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 2, Problem 1: Ornstein–Zernicke Correlations

The Gaussian two-point function is

$$G(x) = \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik \cdot x}}{\gamma k^2 + \mu^2} = \frac{1}{\gamma} \int \frac{d^d k}{(2\pi)^d} \frac{e^{-ik \cdot x}}{k^2 + \xi^{-2}}, \quad \xi^2 = \frac{\gamma}{\mu^2}. \quad (1.1)$$

Explain why it depends only on $r = |x|$. Establish the parameter integral

$$\frac{1}{k^2 + \xi^{-2}} = \int_0^\infty dt e^{-t(k^2 + \xi^{-2})}, \quad (1.2)$$

perform the Gaussian momentum integral, and express the result, up to an overall factor, as

$$G(r) \sim \int_0^\infty dt e^{-S(t)}, \quad S(t) = \frac{r^2}{4t} + \frac{t}{\xi^2} + \frac{d}{2} \log t. \quad (1.3)$$

Use a saddle-point approximation, stating the saddle equation and quadratic prefactor, in the regimes $r \ll \xi$ and $r \gg \xi$. Recover the Ornstein–Zernicke behaviours

$$G(r) \sim r^{-(d-2)} \quad (r \ll \xi), \quad G(r) \sim \frac{e^{-r/\xi}}{r^{(d-1)/2}} \quad (r \gg \xi). \quad (1.4)$$

Solution. Since the free energy respects global $SO(d)$ symmetry, the correlation function must also be invariant under rotation, and hence can only depend on r . The parameter integral is easy to see, and plugging into the correlation function, we find

$$\begin{aligned} G(r) &\sim \int d^d k \int_0^\infty dt e^{-i\mathbf{k} \cdot \mathbf{x} - t(k^2 + \xi^{-2})} \\ &= \int d^d k \int_0^\infty dt e^{-(\sqrt{t}\mathbf{k} + i\mathbf{x}/2\sqrt{t})^2 - t\xi^{-2} - r^2/4t} \\ &\sim \int_0^\infty dt \frac{e^{-t\xi^{-2} - r^2/4t}}{t^{d/2}}. \end{aligned} \quad (1.5)$$

The saddle point is at

$$S'(t_*) = 0 \implies t_* = \frac{r^2}{d \pm \sqrt{d^2 + 4r^2/\xi^2}}. \quad (1.6)$$

Since $t > 0$, we take the plus sign. The correlation function becomes

$$G(r) \sim \int_0^\infty dt e^{-S(t_*) - (t-t_*)^2 S''(t_*)/2} \sim \frac{e^{-S(t_*)}}{\sqrt{S''(t_*)}}. \quad (1.7)$$

For $r \ll \xi$, we have $t_* \sim r^2$, and hence

$$G(r) \sim r^{-(d-2)}. \quad (1.8)$$

For $r \gg \xi$, we have $t_* \sim r\xi$, and hence

$$G(r) \sim \frac{e^{-r/\xi}}{r^{(d-1)/2}}. \quad (1.9)$$

2. Sheet 2, Problem 2: The Two-Point Function as a Green Function

Verify that the correlation function satisfies

$$(-\gamma \nabla^2 + \mu^2) \langle \phi(x) \phi(0) \rangle \sim \delta(x), \quad (2.1)$$

and explain its physical interpretation.

Solution. This is trivial to show. Physically, the correlation function tells us how the system responds to a spiky source of perturbation.

3. Sheet 2, Problem 5*: Scaling in a Lifshitz Theory

Separate $x = (x, \mathbf{y})$ with $\mathbf{y} \in \mathbb{R}^{d-1}$, and consider the free energy

$$F[\phi] = \int d^d x \left[\frac{1}{2} (\partial_x \phi)^2 + \frac{1}{2} (\nabla_y^2 \phi)^2 + \frac{1}{2} \mu_0^2 \phi^2 \right]. \quad (3.1)$$

Write it in Fourier space using momenta $(k, \mathbf{q})$. Under $k' = \zeta k$, $\mathbf{q}' = \zeta^a \mathbf{q}$, and $\phi'_{\mathbf{k}'} = \zeta^{-b} \phi_{\mathbf{k}}$, determine a, b by normalising both gradient terms, and find the running $\mu^2(\zeta)$. In position space, obtain the Gaussian scaling dimension Δ_ϕ . For an interaction $\int d^d x g_n \phi^{2n}$, find the dimension of g_n and show that ϕ^4 is relevant for $d < 7$ and irrelevant for $d > 7$.

Solution.

$$F[\phi_{\mathbf{k}}] = \int \frac{d^d k}{(2\pi)^d} \frac{1}{2} (k^2 + \mathbf{q}^4 + \mu_0^2) \phi_{\mathbf{k}} \phi_{-\mathbf{k}} = \int \frac{d^d k'}{(2\pi)^d} \zeta^{-1-a(d-1)} \frac{1}{2} (\zeta^{-2} k'^2 + \zeta^{-4a} \mathbf{q}'^4 + \mu_0^2) \zeta^{2b} \phi'_{-\mathbf{k}'} \phi'_{\mathbf{k}'}. \quad (3.2)$$

In order to normalise both gradient terms, we must have

$$a = \frac{1}{2}, \quad b = \frac{d+5}{4}. \quad (3.3)$$

Hence, we find

$$\mu^2(\zeta) = \zeta^2 \mu_0^2. \quad (3.4)$$

In position space, we have

$$\phi'(\mathbf{x}') = \int \frac{d^d k'}{(2\pi)^d} \phi'_{\mathbf{k}'} e^{i\mathbf{k}' \cdot \mathbf{x}'} = \zeta^{1+a(d-1)-b} \int \frac{d^d k}{(2\pi)^d} \phi_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} = \zeta^{\Delta_\phi} \phi(\mathbf{x}) \implies \Delta_\phi = \frac{d-3}{4}. \quad (3.5)$$

For an interaction, we have

$$\int d^d x g_n \phi^{2n}(\mathbf{x}) = \int d^d x' \zeta^{1+a(d-1)} g_n \zeta^{-2n\Delta_\phi} \phi'^{2n}(\mathbf{x}') = \int d^d x' g_n(\zeta) \phi'^{2n}(\mathbf{x}'), \quad (3.6)$$

which implies

$$g_n(\zeta) = \zeta^{-n(d-3)/2 + d/2 + 1/2} g_n. \quad (3.7)$$

For $n = 2$, we have

$$g_2(\zeta) = \zeta^{-(d-7)/2} g_2, \quad (3.8)$$

and hence it is relevant for $d < 7$.

4. Sheet 2, Problem 6: Scalar–Gauge Coupling

For a complex scalar and gauge field, use

$$F[\psi, A_i] = \int d^d x \left[\frac{1}{4} F_{ij} F^{ij} + |\partial_i \psi - ie A_i \psi|^2 + \mu^2 |\psi|^2 \right], \quad F_{ij} = \partial_i A_j - \partial_j A_i. \quad (4.1)$$

Find the critical dimension d_c below which the coupling e is relevant and above which it is irrelevant. Discuss what may happen at $d = d_c$. With a term $g|\psi|^4$, the $d = 3$ case is the Ginzburg–Landau theory of a superconductor.

Solution. We have $[A_i] = [\psi] = -(d-2)/2$, and hence $[e] = (d-4)/2$ in length units. Clearly, the critical dimension is $d_c = 4$. Therefore, the coupling is relevant when $d < 4$ and irrelevant when $d > 4$. At $d = 4$ it is marginal at this order; higher-order (loop) corrections determine whether it is marginally relevant or irrelevant.

5. Sheet 2, Problem 7: Emergent Rotational Symmetry

A microscopic cubic lattice in d dimensions has a local order parameter $\phi(x)$ with $\phi \mapsto -\phi$ symmetry. Identify the lowest-order interaction allowed by the lattice rotations but forbidden by full $SO(d)$ rotations. Use scaling to explain why full rotational invariance can emerge at long distances.

Solution. The leading order contributing interaction is

$$\int d^d x \lambda \sum_{i=1}^d \phi \partial_i^4 \phi. \quad (5.1)$$

The coupling constant now has a length dimension $[\lambda] = 2$, which is irrelevant for all d .

6. Sheet 2, Problem 8: Wick's Identity for a Gaussian Ensemble

Let G be a symmetric, positive-definite, invertible $n \times n$ matrix, and set

$$\langle f(\phi) \rangle = \frac{1}{\mathcal{N}} \int_{\mathbb{R}^n} d^n \phi f(\phi) e^{-\phi \cdot G^{-1} \phi / 2}, \quad \mathcal{N} = \det(2\pi G)^{1/2}. \quad (6.1)$$

Prove $\langle \phi_a \phi_b \rangle = G_{ab}$ and, for constant B_a ,

$$\langle e^{B_a \phi_a} \rangle = \exp \left[\frac{1}{2} B_a \langle \phi_a \phi_b \rangle B_b \right]. \quad (6.2)$$

Expand both sides to obtain all three pairings in $\langle \phi_a \phi_b \phi_c \phi_d \rangle$. Show that odd moments vanish and give the general even-moment pairing formula.

Solution. Completing the square gives the generating function

$$Z(B) = \langle e^{B_a \phi_a} \rangle = e^{B_a G_{ab} B_b / 2}. \quad (6.3)$$

Differentiating twice at $B = 0$ gives $\langle \phi_a \phi_b \rangle = G_{ab}$. Four derivatives give

$$\langle \phi_a \phi_b \phi_c \phi_d \rangle = G_{ab} G_{cd} + G_{ac} G_{bd} + G_{ad} G_{bc}. \quad (6.4)$$

Since $Z(B)$ is even in B , all odd moments vanish. Repeated differentiation gives the sum over all pairings for every even moment. This is the finite-dimensional Gaussian version of the contractions in my Tong QFT Problem Sheet 2, Question 9 solution.